# Miguel Masó, PhD

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City: Barcelona (08006) Birth date: 9/4/1989 ORCID: 0000-0003-0962-8550

## Employment history

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| 2022 - today | **SOCOTEC Spain** - Buildings and cities<br>Specialist Engineer, Head of development and innovation |
| 2022 - today | **Universitat Politècnica de Catalunya (UPC)** - Barcelona School of Civil Engineering<br>Adjunct lecturer |
| 2017 - 2022 | **International Centre for Numerical Methods in Engineering (CIMNE)**<br>Doctoral student with FPI grant |

## Education

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| 2017 - 2022 | **PhD in Civil Engineering**<br>UPC, Barcelona, Spain |
| 2013 - 2015 | **Bachelor's degree in Philosophy and Theology**<br>Ponitificia Universtià della Santa Croce, Rome, Italy |
| 2007 - 2013 | **Ingeniero de Caminos, Canales y Puertos** (1st and 2nd cycle studies)<br>UPC, Barcelona, Spain |

## Languages

Native Spanish - Native Catalan - C1 English - B1 Italian

## CV Summary

After studying civil engineering and a period studying philosophy and theology in Rome, he began a doctorate in civil engineering at CIMNE. During the initial years of his doctoral studies, he combined research with work for social purposes at the Montseny Foundation.

Durign the research period at CIMNE, he worked on the stabilization of finite elements for the hyperbolic differential equations, with focus on numerical implementation, multi-physics coupling, and multi-scale resolution. These developments have been succesfully applied to various projects. In addition to the finite element stabilization, he also researched on the particle finite element method, either with moving mesh and fixed mesh.

Currently, he is working at the Building and Cities department of Socotec-Spain. He has participated in several international projects for structural design, taking on responsibilities of vibration assessment and plasticity. He is leading the innovation and development department.

## Publications

### Journal papers

Montanino, Andrea, Miguel Masó Sotomayor, and Alessandro Franci (2025). "One-way coupled Eulerian–Lagrangian strategy for wave propagation and impact on coastal areas". In: *Communications in Nonlinear Science and Numerical Simulation* 147, p. 108845. ISSN: 1007-5704. DOI: https://doi.org/10.1016/j.cnsns.2025.108845.

De-Pouplana, Ignasi et al. (2023). "A prototype of a micro-scale model for the distribution of NO2 in urban areas". In: *Atmospheric Pollution Research* 14.2, p. 101668. ISSN: 1309-1042. DOI: 10.1016/j.apr.2023.101668.

Masó, Miguel, Alessandro Franci, et al. (July 2022). "A Lagrangian-Eulerian procedure for the coupled solution of the Navier-Stokes and shallow water equations for landslide-generated waves". In: *Advanced Modeling and Simulation in Engineering Sciences*, pp. 2213–7467. DOI: 10.1186/s40323-022-00225-9.

Masó, Miguel, Ignasi De-Pouplana, and Eugenio Oñate (Feb. 2022). "A FIC-FEM procedure for the shallow water equations over partially wet domains". In: *Computer Methods in Applied Mechanics and Engineering* 389, p. 114362. DOI: 10.1016/j.cma.2021.114362.

Puigferrat, A. et al. (July 2021). "Semi-Lagrangian formulation for the advection-diffusion-absorption equation". In: *Computer Methods in Applied Mechanics and Engineering* 380, 113807:1–113807:26. DOI: 10.1016/j.cma.2021.113807.

### Theses

Masó, M. (2022). "Coupling shallow water models with three-dimensional models for the study of fluid-structure interaction problems using the particle finite element method". Universitat Politècnica de Catalunya.

### Conferences

Maso, M., A. Franci, et al. (2022a). "A one-way coupled Lagrangian-Eulerian procedure for the solution of landslide-generated waves". In: *WCCM APCOM - 15th World Congress on Computation Mecchanics & 8th Asian Pacific Congress on Computation Mechanics*. Yokohama, Japan (Virtual).

Maso, M., A. Franci, et al. (2022b). "Un Procedimiento Lagrangiano-Euleriano para la Solución Acoplada de las Ecuaciones de Navier-Stokes y Aguas Poco Profundas para Olas Generadas por Deslizamientos de Laderas". In: *CMN XV - Congress on Numerical Methods Engineering*. Las Palmas de Gran Canaria, Spain.

Maso, M., I. De-Pouplana, and E. Oñate (2019). "Coupling shallow water models with 3D models for the study of fluid-structure interaction problems using the particle finite element method". In: *Particles 2019 - VI International Conference on Particle-based Methods*. Barcelona, Spain.

## Stays and other activities

**Guest. Centro de Investigación en Matemáticas.** Purpose: Research. Comparable tasks: WP3 of project TCAiNMaND-PIRSES-GA-2013-612607 Numerical Methods for Real Time Com-

putations Research on stabilized formulations the Finite Element Method applied to hyperbolic conservation laws. 18/07/2017 - 19/10/2017.

## Teaching activities

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| 2022 - today | **Structural Dynamics.** Credits: 7'5. Degree: Master's degree in Structural & Construction Engineering. UPC. |
| 2022 - 2022 | **Structural Analysis.** Credits: 7'5. Degree: Master's degree in Civil Engineering. UPC. |
| 2017 - 2017 | **Numerical Methods.** Credits: 6. Degree: Bachelor's degree in Materials Engineering. UPC. |

## Participation in R+D+i projects

(*2021 SGR 01349*) *Mecànica de Medis Continus i Estructures*
Financing entity: AGAUR. Agència de Gestió d'Ajuts Universitaris i de Recerca
From: 01/01/2022 To: 30/06/2025
PI: Cervera Ruiz, Luis Miguel

(*DPI2015-67857-R*) *Proyectos de I+D: Retos de la Sociedad 2015*
Financing entity: MINECO
From: 01/01/2016 To: 31/12/2018
PI: Codina, Ramon; Baiges, Joan

(*FP7- 612607*) *FP7-PEOPLE-2013-IRSES*
Financing entity: EC
From: 01/01/2014 To: 31/12/2017
PI: Larese de Tetto, Antonia

## Miscellaneous

- **Skills:** Computational Mechanics, Fire Engineering, Ocean Engineering, Vibration Assessment
- **Programming:** C++, Python, C#, .NET
- **Tools:** Robot Structural Analysis API, Revit API
- **Miscellaneous:** ENVISION SP 2023-2024